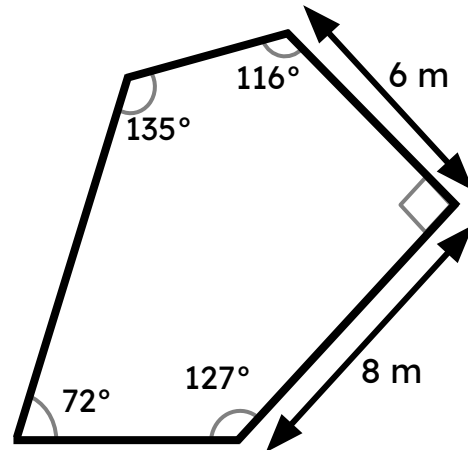
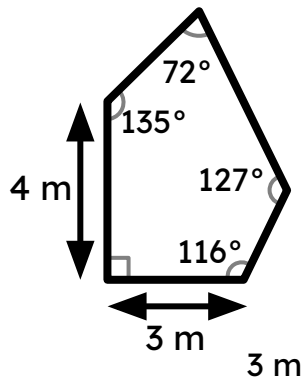




Similarity in shapes

Task A: Identifying whether shapes are similar

1) Are the two shapes below similar? Justify your answer with reasoning.



Shape A's angles: 72° , 127° , 116° , 90° , 135°

Shape B's angles: 72° , 127° , 90° , 116° , 135°

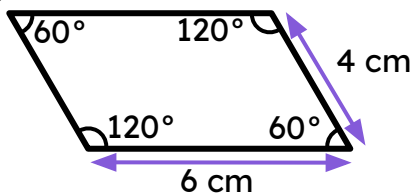
The angles are not in the same order, so the shapes are not similar.

2) Alex, Izzy, Jun and Sam have drawn some quadrilaterals.



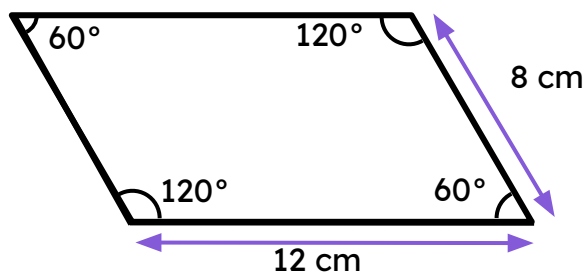
Alex

This is my quadrilateral.



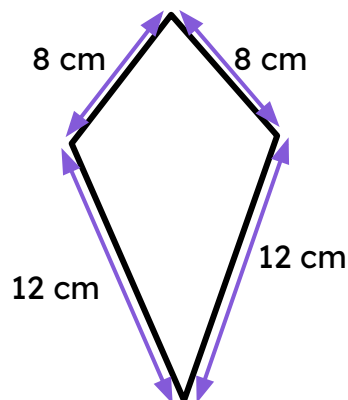
Izzy

My quadrilateral is similar to Alex's and all of the edges are twice the length.



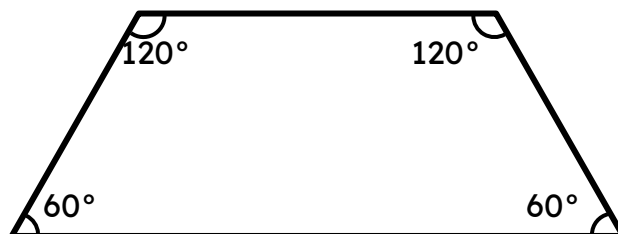
Jun

My quadrilateral is not similar to Alex's, even though all the edges twice the length.



Sam

My quadrilateral is not similar to Alex's, even all the angles are the same.

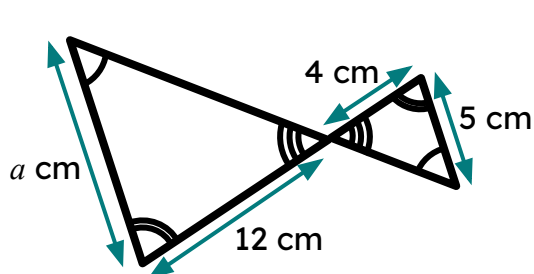


Sketch a possible quadrilateral for Izzy, Jun and Sam.

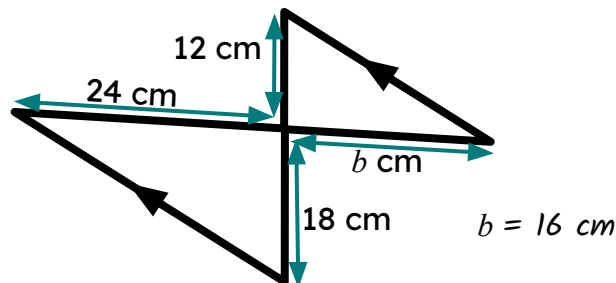


Task B: Finding similar shapes in complex diagrams

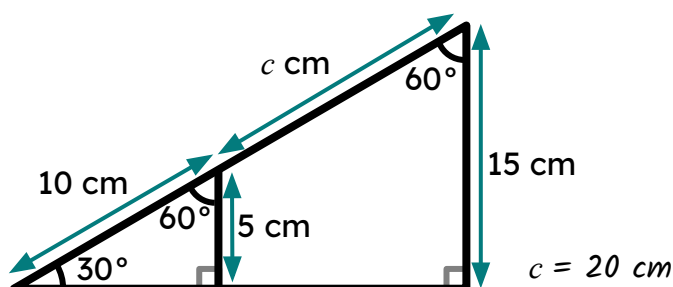
1) Find the value of each unknown.



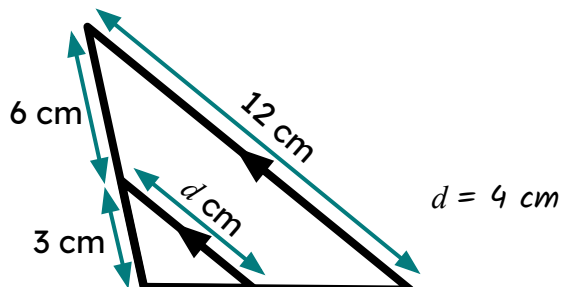
$$a = 15 \text{ cm}$$



$$b = 16 \text{ cm}$$

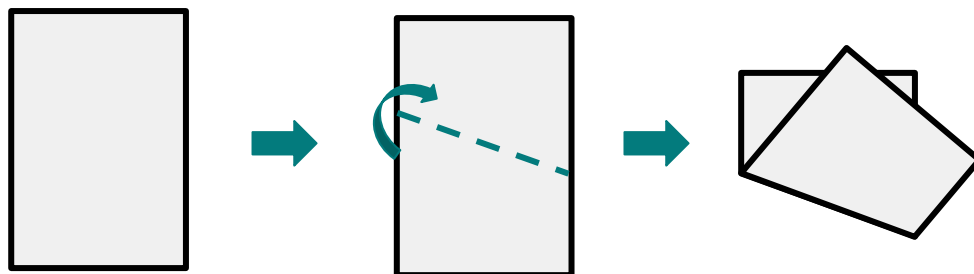


$$c = 20 \text{ cm}$$

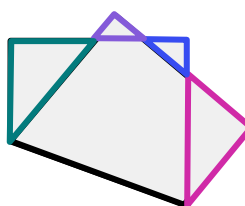


$$d = 4 \text{ cm}$$

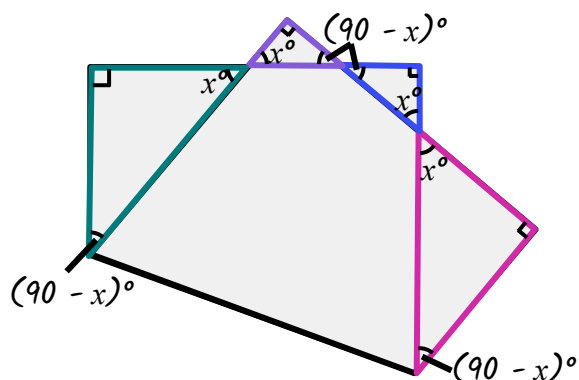
2) Take a piece of paper and fold it, as shown below.



The paper that does not overlap makes four triangles.



Prove that these four triangles are all similar.



Each triangle contains a right angle from the four corners of the paper. Let one angle be x°

Angles in a triangle sum to 180° .

$$180 - 90 - x \equiv 90 - x$$

Vertically opposite angles are equal and angles in a triangle sum to 180° . $180 - 90 - (90 - x) \equiv x$

All four triangles contain the same angles, so they are all similar.

